

Kubernetes Syllabus

Unit-1: Introduction Kubernetes

- Introduction to Kubernetes
- Kubernetes Cluster Architecture
- Core Concept of Kubernetes Services
- ETCD & Controller & API & scheduler
- Exploring your Cluster
- Understanding YAML

Unit-2: Installation

- Installation minikube
- Installation using kubeadm utility in HA mode
- Install Kubernetes the hard Way
- How to use kops and create
- Kubernetes cluster

Unit-3: Application Deployment

- Creating a Deployment in Kubernetes using YAML
- Creating a Service in Kubernetes
- Understanding about pod & Replication & Deployment configuration
- Using Rolling Updates in Kubernetes
- Configure Environment variable in application
- Configure secret resource for sensitive value
- Creating Config Map
- Scale Applications
- Multi Container PODs
- Init Containers
- Self-Healing Applications

Unit-4 Storage management

- Creating Persistent volume
- Persistent Volume Claim
- Volume claim policy understanding
- Attach storage on deployment

Unit-5 Pod Scheduling

- Manual Scheduling
- Labels and Selectors
- Taints and Tolerations
- Node Selectors
- Node Affinity

- DaemonSets
- Static Pods
- Multiple Schedulers
- Configuring Kubernetes Scheduler

Unit-6 Resource Allocation

- Restrict Limit Memory & CPU use
- Creating Resource Quota
- Creating Limit Quota

Unit-7 Networking

- Network Namespace understanding
- Docker Network
- Deploy Kubernetes Network
- Cluster Networking
- Pod Networking
- Creating Service Network
- DNS Concept in kubernetes
- Ingress System kubernetes

Unit-8 Monitoring Kubernetes

- Logging and Monitoring
- Monitoring Cluster Component
- Managing application Logs

Unit-9 Security

- Authentication
- TLS Introduction
- Certificate System kubernetes
- Creating Certificate
- Role base Access Controls
- Cluster Role and Role Binding
- SCC: Security Constant Conta & Network Policy
- Image Security

Venue Details

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